

Question Paper

Exam Date & Time: 21-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: PHY 1071/PHY_1071 ENGINEERING PHYSICS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) With necessary diagram, obtain an expression for angle of acceptance and numerical aperture for an optical fiber. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) A 0.50 kg baseball is confined between two rigid walls of a stadium that can be modeled as a "box" of length 100 m. Calculate the minimum speed of the baseball. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1C) State whether the given statement is true or false and justify. "Germanium is more suitable material than Silicon for single junction solar cell". (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2A) Write the assumptions made in Planck's hypothesis of blackbody radiation. How Planck's law succeeds in explaining the short and long wavelength regions of radiation. (5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) A 30-eV electron is incident on a square barrier of height 40 eV. What is the probability that the electron tunnels through the barrier if its width is 0.1 nm? And how does this change when the width is doubled, and the energy is halved? (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) State whether the given statement is True or False and justify. Energy levels are equally spaced in simple harmonic oscillator. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Assuming the Fermi-Dirac distribution function, obtain an expression for the Fermi energy E_F in a metal at zero K. [Given: density-of-states function, (5)

$$g(E) dE = \frac{8\sqrt{2} \pi m^{\frac{3}{2}}}{h^3} E^{\frac{1}{2}} dE]$$

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) The $J = 0$ to $J = 1$ rotational transition of the CO molecule occurs at a frequency of 1.15×10^{11} Hz. Use this information to calculate the moment of inertia and bond-length of the molecule. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) The superconductors are perfect diamagnets. Justify this statement. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) Explain why ferroelectric material is considered as the non-linear dielectrics with a schematic diagram. Describe two main characteristics of ferroelectric substances. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) A dielectric material has $\epsilon_r = 4.94$ and $n^2 = 2.69$. Calculate the ratio between electronic and ionic polarizability of this material. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) State whether the given statement is true or false and justify. Type-II superconductors have been used in strong field superconducting magnet over Type-I superconductors'. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Explain how the surface to volume ratio influences the different properties of materials with examples. (5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Explain top-down and bottom-up approach for preparation of nanomaterial using schematic diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) State whether the given statement is true or false and justify. 'Emission colour of the quantum dot are constant with the change in its size' (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 PHY_1071:ENGINEERING PHYSICS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	22
4	4	CO 4	8
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 PHY_1071:ENGINEERING PHYSICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	25(50%)
3	Application	9	23(46%)
4	Analysis	1	2(4%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)